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Advancing Language Assessment with GPT: Is It Nonnative-Language Friendly?

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ABSTRACT
Despite the growing interest in AI for language assessment, there remains a significant research gap regarding its usefulness for assessing *less* proficient language skills, particularly those of learners of English as a second or foreign language (S/FL). AI models often prioritize proficient writing, neglecting the intricacies of learner language. This study addresses this gap by evaluating the capacity of AI to replicate the written performances of both L1 and L2 novice academic writers, with a focus on formulaic language. Using an AI tool based on the GPT-2 model, two AI-generated essay corpora were created to closely resemble existing L1- and L2-English corpora of argumentative essays by first-year university students. The comparative analysis reveals AI's inability to accurately replicate S/FL learner language. Notably, the AI model failed to generate typical "learner bundles," which are formulaic language patterns considered unique to or characteristic of English S/FL learners. The findings of this study indicate an urgent need for improving AI technology to more effectively represent the diverse linguistic attributes of S/FL learner language, an improvement which may play a critical role in fair and effective language assessment.

INTRODUCTION

While research on artificial intelligence (AI) as a language-learning medium is flourishing, recent studies have clustered around spoken language practices, such as the effects of interactive chatbots in English educational settings (e.g., Hsu et al., 2021; Jeon, 2022; Smutny & Schreiberova, 2020; Yang et al., 2022). Meanwhile, relatively few efforts have been made to examine AI-based tools for teaching or assessing academic written language, with most research on this topic focusing on automated essay scoring (AES) systems. These systems, driven by natural language processing algorithms, provide diagnostic scores and comprehensive feedback on written performances (e.g., McCarthy et al., 2022; Ragheb Al-Ghezi et al., 2023; Ranalli, 2021). Some research on second language (L2) academic essays has scrutinized the reliability of AES by contrasting it with human ratings (e.g., Shermis, 2014; Vo et al., 2023; Weigle, 2013). Despite advances in AES systems, challenges still exist for their practical application, especially in dealing with subgroups from diverse language backgrounds, such as learners of English as a second or foreign language (S/FL). Unique writing styles associated with these backgrounds have raised concerns about fairness in the assessment of high-stakes writing tasks, as AES systems' scores and those of human raters

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do not always agree (e.g., Breyer et al., 2017; Bridgeman & Ramineni, 2017; Vo et al., 2023). The discrepancies are, at least in part, linked to AES systems' reliance on proficient English writing standards, a fact which can result in lower scores for written performances that do not adhere to first language (L1) English norms (e.g., Bridgeman & Ramineni, 2017; Vo et al., 2023). Consequently, the effective integration of AI into language assessment hinges on its capability to grasp the subtleties of learner language (e.g., Hannah et al., 2023). Addressing this issue requires using learner data to train AES systems, as this approach has been shown to improve scoring reliability and better reflect the characteristics of S/FL performance (e.g., Xu, 2018; Xu et al., 2021).

Building on the discussion of assessment challenges in L2 writing and the relevance of AI in language assessment, recent attention has been directed toward GPT language models developed by OpenAI (e.g., Xi, 2023). These models, including ChatGPT, released in November 2022, have garnered significant interest. While much of the initial focus has been on their potential for generating human-like conversational responses, their capacity to produce academic writing remains underexplored. This is particularly crucial for assessing whether AI-generated texts can replicate the specific features of novice writers, especially those from diverse linguistic backgrounds. Moreover, the ability of GPT models to reflect the structural and functional aspects of learner language – particularly in written form – has not been thoroughly investigated. To address this gap, the current study compares the written performances of incoming college students, both native English speakers (L1-English) and nonnative English speakers (L1-Korean), with written performances generated by a GPT language model on the same topics. As an evaluative measure, the study uses recurrent multiword sequences, or lexical bundles, to examine argumentative essays produced by novice academic writers and AI. This exploration into AI's ability to produce writing styles resembling those of specific human populations holds considerable promise, as it could significantly advance language learning and assessment practices.

Given the profound implications of integrating AI into educational contexts, it is critical to align such advancements with ethical considerations. As AI gains traction in education and assessment, research exploring its uses in these contexts has a particular obligation to consider ethical implications. It is no secret that AI can reinforce societal biases, and it is evident that the integration of AI tools into language education will shape how students engage with both language and content, including cultural content. Therefore, educational researchers and practitioners must be prepared to address and manage AI's potential negative effects, as well as striving to ensure that the advantages of AI are available to all learners. Consequently, our exploration includes a reflective examination of these ethical dimensions to promote responsible usage of AI in educational assessment.

LITERATURE REVIEW

Lexical bundles used by native and nonnative writers

Research on formulaic language has demonstrated its substantial importance in natural discourse (e.g., Hyland, 2012; Qin, 2014; Wray, 2006). In the context of second/foreign language education, formulaic language has been studied in relation to the developmental trajectory of learning. Because smooth social interaction involves many routine sequences, and the expected formulaic language varies from context to context, its use can be correlated with competence in communication, as Pawley and Syder (1983) were among the first to argue. Because formulaic sequences may be learned and produced as "chunks," they are also crucial to effective language processing and fluency, as well as comprehension (e.g., Nekrasova, 2009; Tremblay et al., 2011; Wood, 2015). In addition, studies have shown that such language is used in particular ways in specific registers, such as academic prose (e.g., Appel, 2022; Hyland, 2012; Shin & Won, 2024; Wright, 2019). Hence, assessing the use of formulaic sequences can serve as a means of tracking S/FL learners' language development.

Some data-driven formulaic language research has focused on lexical bundles (LBs), which are defined by frequency: LBs are combinations of three or more words that frequently appear together in a specific register (Biber et al., 1999). This type of formulaic sequence has been examined in different types of academic prose such as published journal articles and student essays (e.g., Li et al., 2020; Lu & Deng, 2019; Wright, 2019). The findings of this line of research indicate that academic writing in English commonly contains certain bundles (e.g., De Cock & Granger, 2021; Pérez-Llantada, 2014; Shin & Won, 2024; Wright, 2019) and that the presence of these bundles marks academic fluency (e.g., Appel, 2022; Bychkovska & Lee, 2017; Li et al., 2020). Because specific bundles are expected in specific genres, using appropriate bundles helps writers convey information in contextually appropriate ways, as well as creating textual coherence (e.g., Hyland, 2012; Li et al., 2020; Northbrook et al., 2022). Therefore, the mastery of the use of this type of formulaic language is required, or at least expected, for disciplinary membership (e.g., Kim & Kessler, 2022; Lu & Deng, 2019; Wright, 2019).

Research on LBs in the academic written production of L1 and L2 writers of English has observed differences in the features of these groups' written discourse (e.g., Ädel & Erman, 2012; Bychkovska & Lee, 2017; Dahunsi & Ewata, 2022; Paquot, 2017). For example, Chen and Baker (2010) compared L1-English and L1-Chinese English speakers' LB usage. In a quantitative analysis of three corpora-one of published academic texts, one of L1 student academic writing, and one of L2 student academic writing – the study found more clausal bundles in the writing of both groups of student writers and more phrasal bundles in the published texts. In addition, the authors conducted a qualitative analysis to investigate concordance lines, examining the larger contexts of the frequently occurring bundles and observing several features particular to the L2 students' writing. Specifically, the L2 writers relied heavily on a small set of expressions that barely appeared in the published texts, while rarely using the bundles that appeared most frequently in the two L1 corpora (i.e., the L1 student writing and the L1 published writing). Furthermore, both groups of L1 writers skillfully employed diverse hedging strategies, thus expressing their arguments and conclusions with caution, while the L2 writers relied on fewer hedges, leading to an impression of more starkly declared arguments and conclusions. The authors did point out, however, that the features distinguishing L2 writers diminished with L2 proficiency, with L1 and higher-proficiency L2 writing becoming more similar. Several other L2 studies have reported similar findings (e.g., Ädel & Erman, 2012; Dahunsi & Ewata, 2022; Pérez-Llantada, 2014).

In contrast, Shin's (2019) study comparing the use of LBs by L1-English and L1-Korean EFL undergraduate writers noted little difference, observing that the two groups produced writing that shared features characteristic of novice writers, including the absence of LBs that have been documented in skilled academic writing. This finding underscores the fact that academic writing conventions must be learned not only by nonnative speakers but also by native speakers and supports previous research claims that the complexity of academic writing grows with writers' language development (e.g., Biber & Gray, 2010; Biber et al., 2011; Pan et al., 2016).

AI text generation research

Text generation is an integral part of natural language processing (NLP), with broad applications in machine translation, dialog systems, and news generation (e.g., Hochreiter & Schmidhuber, 1997; Luong et al., 2015; Rumelhart et al., 1986). The transformer model, an AI development, has revolutionized the field of NLP (Vaswani et al., 2017). It is designed to understand and learn the relationships within input data through a self-attention technique. Subsequent advances in NLP technologies include the Bidirectional Encoder Representations from Transformers (BERT), aimed at sentence identification, and the Generative Pre-trained Transformer (GPT), designed for new sentence generation (Devlin et al., 2019; Radford, Narasimhan, et al., 2018). Recent research has strived to develop pretrained language models that comprehend language; text generation based on these models has reached a high level of proficiency that parallels human writing (e.g., Radford, Narasimhan, et al., 2018).

The initial GPT model employed the BooksCorpus, consisting of over 7,000 unique unpublished books across various genres, as its training dataset (Radford, Narasimhan, et al., 2018). This corpus, with its long contiguous texts, conditions the generative model to handle cohesive information. The GPT-2 model was subsequently trained on a large corpus of eight million documents (totaling 40 GB of text) known as WebText, generated by scraping pages linked to Reddit (a social media platform) posts that had received at least three upvotes (Radford, Wu, et al., 2018). A count of three or more upvotes is considered an indication that other users found the link valuable (e.g., interesting, educational). These advancements enabled GPT models to generate highly proficient texts, closely resembling native speaker output across various domains. More recent iterations, such as GPT-3 and GPT-4, have further enhanced these capabilities, enabling even more sophisticated language production (Brown et al., 2020; Open et al., 2023). However, as these models become more adept at generating polished writing, their ability to replicate less advanced, learner-like texts remains largely underexplored.

Most existing studies focus on assessing AI-generated proficient writing across genres such as poetry (Köbis & Mossink, 2021), story writing (Fang et al., 2023), and case reports (Pinto et al., 2024), with a particular concentration on academic research articles (e.g., Awosanya et al., 2024; Dergaa et al., 2023; Kacena et al., 2024; Salvagno et al., 2023; Semrl et al., 2023; Tang et al., 2024). In contrast, there remains a scarcity of research exploring the potential of AI to simulate less proficient writing from individuals with diverse language backgrounds, including English S/FL learners. Little is known about whether AI can effectively model learner language by capturing the unique patterns and features exhibited by students across different proficiency levels. Exploring this gap is critical, as it could illuminate how well AI-generated texts mimic the writing of less advanced learners and the extent to which AI can be leveraged as a tool for language assessment. A close scrutiny of this aspect could provide valuable insights into the differences between machine- and human-produced written performances, thereby informing how writing assessment and learning should be done in an era in which students will increasingly seek help from various AI models in producing their written tasks.

THE PRESENT STUDY

This study explores the extent to which AI texts reflects the features characteristic of texts produced by novice academic writers from different language backgrounds. Two AI text corpora were generated using GPT-2, which was trained on two existing corpora of academic essays written by L1- and L2-English first-year university students. The four corpora are strictly matched in terms of genre (i.e., argumentation) and topic (see Appendix A). As an evaluative measure, this study employs the frequency-based formulaic sequences known as LBs. It identifies and compares the frequently occurring LBs in each of the four corpora (i.e., L1 essays, AI-L1 essays, L2 essays, and AI-L2 essays). The structures and discourse functions of the most frequent LBs are then analyzed and compared. Due to the integral role of LBs in discourse, the findings of this study will illuminate the extent to which the essays generated by AI resemble those produced by novice academic writers at the university level. To this end, the following two research questions are posed:

(1) In matched corpora of argumentative essays, what are the most frequent four-word LBs in the student texts and in the AI texts?
(2) Do the LBs identified in the student texts and those in the AI texts differ with respect to structural and functional types?

METHOD

Corpus data description

This study used human-generated data in existing L1 and L2 corpora (Shin, 2018; Yoo & Shin, 2022) as a basis to develop an AI text dataset specifically for the study. First, the L2 corpus data, collected at a South Korean university, consist of argumentative essays assigned as part of an English language course placement test for incoming freshmen. These essays were written by L1-Korean students with varying degrees of English proficiency. The L2 corpus contains texts (average length = 268.1 words; total words just over 1 million) written by a total of 3,958 students. The L1 corpus was designed to match the L2 corpus. Compiled at a university in the United States, the L1 corpus consists of argumentative essays assigned as a diagnostic test during the first week of a composition course for incoming freshmen. These writers' language skills were expected to be typical of native English speakers at the onset of university-level studies; they were not expected to be familiar with advanced academic English prose conventions. The L1 corpus contains texts (average length = 358.1 words; total words = 363,131) written by 1,014 native-English-speaking students.

The study applied AI to texts from established datasets, creating a direct link between the raw data input and its synthetic augmentation. The L1 and L2 student corpora were utilized as the source data for the GPT-2 model to generate essays for the AI-L1 and AI-L2 corpora, respectively. The AI-L1 corpus was generated based on L1 student essays, while the AI-L2 corpus was generated based on L2 student essays. The process we used applied the "beam search" technique to each sentence. Beam search is a method commonly used in sentence generation, which considers a fixed number of the most probable next steps at each point in the sequence. For this study, we selected the three most likely next words at each step, using a beam width of three. This approach allows the algorithm to track the three best options at each stage before proceeding, effectively balancing exploration and exploitation during sentence generation. In addition, we set the probability distribution threshold at 0.94 for this process. To foster creativity, we selected 100-word sets and set the creativity rate at 0.8.

To avoid repetitive use of any word or phrase, we implemented a repeat penalty of 1.5. Finally, to ensure coherence, we allowed for more tokens (i.e., sentences) to be generated and for the process to produce longer essays by making the number of tokens flexible. This method ensured that each generated sentence was contextually connected to the previous one, leading to more cohesive and natural output.

The process of sentence generation, which involved four main algorithms, was as follows.[1]

(1) Input sentences were selected randomly within a minimum range of the training essay data, which comprised datasets of argumentative essays written by L1- and L2-English-speaking university students. To maintain a natural context, the input sentences were not shuffled, and the minimum and maximum range was flexible, adjusting according to the length of the input essay data, allowing up to four sentences to be input.
(2) "Masking," where parts of the input are randomly hidden (i.e., masked) from the AI tool, was applied in order to generate new sentences, including those that differed in structure from any in the input. When portions of the input are masked, the AI fills the gaps. We used this masking strategy in an attempt to train the model to handle diverse linguistic structures and contexts.
(3) The pretrained GPT-2 model was used to generate new essays by inputting the first few phrases/sentences from the L1 essays for the AI-L1 dataset, or from the L2 essays for the AI-L2 dataset, and then allowing the model to generate text.
(4) The generated essays were evaluated by comparing them to the essays in the training dataset and assessing their similarity in terms of grammatical structures and overall coherence.
(5) After the generated essays were assessed, we made further adjustments and fine-tuned the model to improve its performance. Among the adjustments were changes to the parameters (number of beams, probability distribution threshold, number of potential words, creativity rate, and masking ratio) and changes to the chosen input data. Each time we made such adjustments and fine-tuned the model, steps 3–5 were repeated. Every new set of AI texts was compared with the input data until we were satisfied with the AI's output. We operationalized "satisfactory performance" to capture our goal not of creating perfect matches but of producing AI-generated essays that were similar to the human-generated essays in terms of language-based assessment criteria, including the complexity of grammatical structures, lexical choice, appropriacy to context, and discourse coherence.

[1]See Appendix B for more details on each algorithm.

Figure 1 illustrates the practical application of this process by presenting a pair of essays: one written by an L2 student and its AI text counterpart, both addressing the topic of comparing book knowledge and experience. As explained above, the AI text was created by using the first few phrases and/or sentences from the L2 student essay as input. The parts highlighted in red are identical in both essays, although not necessarily in the same position.

Table 1 summarizes the four corpora, including information on the data according to topic (see Appendix A): L1 student essays (L1), AI-generated essays based on L1 student essays (AI-L1), L2 student essays (L2), and AI-generated essays based on L2 student essays (AI-L2).

The AI corpora were created to match the respective human-generated corpora as closely as possible in overall size (i.e., total number of words) as well as average essay length, although the match was not perfect. The L1 corpus and the AI-L1 corpus include the same number of L1 essays ($n = 1{,}014$) of similar lengths. However, the AI-L2 corpus contains a smaller number ($n = 3{,}699$) of longer texts than the L2 corpus (320.2 words vs. 256.7 words). The number of total words in each of the four corpora differs considerably, but the total numbers of words by topic in each of the two corpora that the study compares – L1 versus AI-L1, and L2 versus AI-L2—are similar, a fact which allows for valid comparisons to be made when we identify LBs.

Identification and analysis of lexical bundles

To address the first research question, separate sets of LBs were identified in the four corpora (i.e., L1, AI-L1, L2, and AI-L2) for each of the four essay topics using *AntConc* (Anthony, 2023). For the present study, the frequency threshold was set at 20 times per about half a million words, with the threshold adjusted according to the size of each dataset by topic; the range threshold is a minimum of five different texts across the data. Two kinds of LBs were removed from the final datasets: any exact quote from the essay prompts and any overlapping four-word sequence from the same longer sequence.

To address the second research question, the identified LBs were categorized according to their structures and functions, following a taxonomy proposed by Biber et al. (2004). In terms of structure, there are two main types: (a) clausal, or verb phrase (VP)-based; and (b) phrasal, or noun phrase (NP)- or prepositional phrase (PP)-based. Clausal bundles contain a verb, for example, *there are lots of*. NP-based bundles begin with a noun phrase that may be followed by an *of*-phrase fragment, as in *the rest of the*, or a postmodifier fragment, as in *students who are interested*. PP-based bundles begin with a preposition, which is followed by a noun phrase fragment, for example, *in the middle of*.

In terms of the LBs' functions, there are three main types based on the LBs' role in the discourse: (a) stance expressions, which convey the speakers' attitude toward a proposition, such as *it would be the*; (b) discourse organizers, such as *at the same time*; and (c) referential expressions, such as *is one of the*. If an LB could be interpreted as having multiple discourse functions, it was classed based on the function that it served most frequently in the dataset (e.g., Pan et al., 2016).

[L2 student essay]

Some people might say that knowledge gained from books is more beneficial than knowledge gained from experience. However, I think that knowledge gained from experience is more important than the other one. **These two types of knowledge have quite different features. Commonly, these two types of knowledge** can both help people to learn something about they don't know. However, they are not all the same because they are gained from different sources that have ultimately distinctive origins.

Firstly, knowledge gained from experience is created through the subjects' cognitive procedure. It is not gained from natural and unconscious progress, but it gained from person's true effort to learn about something. Because of this feature, knowledge gained from experience is more long-lasting than knowledge gained from books. Although knowledge included in books is related with numerous fields, it is ultimately limited knowledge source because people can not examine it in person. So, it is easily forgotten by people. For example, when I first learn about computer programming, I didn't understand and memorize the detailed procedure by just studying computer related books. However, through real experience with computer, I could finally memorize them until now.

Secondly, knowledge gained from experience is trustful than knowledge gained from books. In recent days, as there are lots of advances in technology, social science and humanity, books about lots of subjects have been published. Although some books written by good writers have profound knowledge that is real, there are also some of the books that is written by people that only have superficial knowledge about the subjects just for fame and fun. The knowledge that these kind of books contain is not always true and sometimes it is fake. For common public, it is not easy to distinguish the good-quality books with the bad-quality books. Therefore. in my opinion, it is better to get profound knowledge through their own effortful experience about interesting subjects with professors or other prominent figures not with just books.

For these reasons, I think knowledge gained from experience is more beneficial and important source than knowledge gained from books.

[AI-generated essay developed following the L2 student essay above]

Some people might say that knowledge gained from books is more beneficial than knowledge gained from experience. However, I think that knowledge gained from experience is much more valuable.

The reason for this is that experience teaches you how to deal with situations that you might not be able to handle on your own. For example, if you have never been in a car accident, you probably won't know what to do when you are in one. If you've never had a fight with your girlfriend, then you will probably have a hard time dealing with it when it happens.

On the other hand, books can teach you a lot about a subject, but they can't tell you everything about it. You have to go out into the real world and find out for yourself what works and what doesn't work. This is why I believe that books are not the best way to learn a new skill, because they don't give you all the information you need to know to get the job done. **These two types of knowledge have quite different features. Commonly, these two types of knowledge** are referred to as practical and theoretical knowledge. Practical knowledge is what you use in your everyday life. It is the knowledge you can use to solve everyday problems, such as finding a parking space, getting to work on time, or getting a job interview. Theoretical knowledge is knowledge that is used in the study of a particular subject. A good example of this would be the theory of quantum mechanics. Quantum mechanics is a branch of physics that deals with the properties of subatomic particles and how they interact with each other and the world around them. There are many different theories about how the universe works, and each theory has its own strengths and weaknesses. Some theories are more useful than others, so it is important to understand which theory is right for you before you decide which one to use.

Figure 1. Sample essays from an L2 student and from AI.

Table 1. Description of the corpus data.

	L1	AI-L1	L2	AI-L2
TOPIC 1				
Number of essays	158	158	891	858
Total size (words)	56,369	54,352	227,895	257,529
TOPIC 2				
Number of essays	237	237	1,701	1,506
Total size (words)	86,123	92,191	457,829	482,950
TOPIC 3				
Number of essays	448	448	771	755
Total size (words)	158,448	154,998	186,696	257,642
TOPIC 4				
Number of essays	171	171	595	580
Total size (words)	62,191	65,395	143,741	186,196
TOTAL				
Number of essays	1,014	1,014	3,958	3,699
Mean length of essays	358.1	361.9	256.7	320.2
Total size (words)	363,131	366,936	1,016,161	1,184,317

RESULTS

Lexical bundles by essay topic

Table 2 presents lists of LBs identified in the four corpora. Due to limits of space, only the first 56 bundles are shown in the table. The L1 corpus contains 380 LBs in total (Topic 1: 86, Topic 2: 97, Topic 3: 91, Topic 4: 106), while the AI-L1 corpus contains 1,348 in total (Topic 1: 391, Topic 2: 381, Topic 3: 306, Topic 4: 270). The L2 corpus contains 391 bundles in total (Topic 1: 95, Topic 2: 100, Topic 3: 102, Topic 4: 94), while the AI-L2 corpus contains 1,480 in total (Topic 1: 378, Topic 2: 366, Topic 3: 396, Topic 4: 340). Recall that the AI text corpora are based on existing student essays and consist of responses to the same essay prompts, resulting in total sizes that are quite similar. However, the numbers of bundles are significantly different, with both human-generated corpora containing fewer than 400 bundles, while both AI text corpora contain over 1,300 bundles. In addition, in the two AI text corpora, the frequency and range of bundles are almost always the same, such that each bundle tends not to occur more than twice per essay, in contrast to the L1 and L2 student essays, which tend to rely heavily on a few specific bundles. In short, it appears that AI texts draw on relatively large repertoires of formulaic language, compared with human-generated texts.

Grammatical structures of LBs by topic

L1 essays and AI-L1 essays

Figures 2 and 3 show the proportions of the three main structural types (NP-based, PP-based, and VP-based) of LBs identified in the L1 corpus and the AI-L1 corpus by topic. Across topics and across the two corpora, VP-based bundles were the most frequent. Of the two kinds of phrasal bundles, NP-based bundles were used slightly more than PP-based bundles in AI-L1 essays on all four topics and in L1 essays on Topics 1 and 2, while they were used at similar rates in L1 essays on Topics 3 and 4.

Table 3 presents the proportions of structural subcategories, which are strikingly similar in the two corpora.

Table 2. LBs in human-generated (L1, L2) and ai-generated (AI-L1, AI-L2) essays across topics.

L1		AI-L1		L2		AI-L2	
on the other hand	67	if you want to	121	I would like to	206	if you want to	295
is more important than	54	**it is important to**	64	so I want to	200	are a lot of	285
when it comes to	45	**on the other hand**	56	so if I could	163	**I would like to**	239
it would be the	37	in a way that	55	I think it is	139	**for a long time**	187
the best way to	35	**the best way to**	52	**the most important thing**	119	**there are so many**	176
thing I would change	29	**to be able to**	48	people who live in	115	one important thing about	155
in my opinion I	26	was born and raised	46	**is one of the**	114	I was born and	147
to be able to	25	**when it comes to**	45	**there are so many**	86	to be able to	140
with the statement that	19	**not be able to**	45	**there are a lot**	82	is more important than	136
there is a lot	18	**will be able to**	44	why I want to	72	**have a lot of**	136
know what to do	18	this is why it	44	**there are lots of**	68	so I decided to	131
I was born in	18	**the only way to**	44	one of the most	63	the best way to	117
I grew up in	18	no such thing as	44	that I want to	62	what do you think	115
not be able to	17	**I grew up in**	44	**a lot of people**	61	to live in a	113
there are some things	16	you are interested in	43	to solve this problem	60	I think it would	108
out of a book	16	**a great way to**	42	my hometown is very	60	to go back to	105
learn from each other	16	**you need to know**	41	so if I can	54	no such thing as	105
I would like to	16	**the world around us**	39	what I want to	53	**to go to the**	103
book on how to	16	why it is so	37	**on the other hand**	53	lot of people who	101
learning from a book	15	a lot of books	36	to change about my	52	**is one of the**	100
is one of the	15	**a better understanding of**	36	there are too many	49	**on the other hand**	99
from reading a book	15	to become a better	33	when I go to	48	**a lot of things**	99
you are able to	14	there are many other	33	I was born in	46	**the most important thing**	97
to learn how to	14	the most important thing	32	I love my hometown	46	if you don't know	94
one of the most	14	same is true for	32	I can change my	46	if you have any	91
know how to use	14	a lot of things	32	system in my hometown	45	to find a place	90
go hand in hand	14	a great place to	32	there are many things	43	in a way that	89
due to the fact	14	to share with you	31	in my high school	43	**there are a lot**	87
by reading a book	14	the world around you	30	**has a lot of**	43	**will be able to**	85
at my high school	13	**by reading a book**	30	living in my hometown	42	the first time I	85
a better understanding of	13	what is going on	29	when I was young	40	I don't know if	83
we are able to	12	in a small town	29	**to go to the**	40	the same is true	80
to the fact that	12	if you have any	28	a chance to change	39	I don't know what	78
to ride a bike	12	**I would like to**	28	**there are many people**	38	I'm not sure if	75
to read a book	12	they are not the	27	**have a lot of**	38	I feel like I	75
the most important source	12	there are a lot	27	for these reasons I	38	all over the world	75
my hometown is the	12	in the middle of	27	first of all I	38	not sure if this	74
in many different ways	12	are a lot of	27	**a lot of money**	38	it would be better	74
about my high school	12	it is also important	26	I agree with the	37	it would be a	74
will be able to	11	I am going to	26	therefore I want to	36	**there are lots of**	72
there are so many	11	**there are so many**	25	I really want to	35	in a place where	72
there are many things	11	**one of the most**	25	so I think that	34	when it comes to	71
the world around us	11	in the same way	25	is much more important	34	**has a lot of**	71
people are able to	11	I am not saying	25	to go to school	33	a good idea to	71
in the real world	11	over and over again	24	my hometown has many	33	not be able to	69
in my opinion the	11	for a long time	24	it is hard to	33	with a population of	67
have a lot to	11	to know how to	23	if I have a	33	to make sure that	67
don't get me wrong	11	there are a number	23	I have to go	33	to take care of	65
books are a great	11	is more important than	23	are a lot of	33	to go to a	65
you need to know	10	will give you a	22	when I was in	32	make it easier for	65

(*Continued*)

Table 2. (Continued).

L1		AI-L1		L2		AI-L2	
would have to be	10	to make sure that	22	for example there are	32	I think it's a	64
they are able to	10	so I decided to	22	but if I could	32	**but I think it**	64
tell you how to	10	it is so important	22	one of the biggest	31	won't be able to	63
is the most important	10	if you do not	22	**for a long time**	31	no choice but to	63
is the best teacher	10	feel free to leave	21	I would make a	30	if you are a	63
how to do things	10	as a result of	21	I am sure that	30	a good place to	63

Note: LBs that occur in both the L1 and AI-L1 corpora or in both the L2 and AI-L2 corpora are in bold.

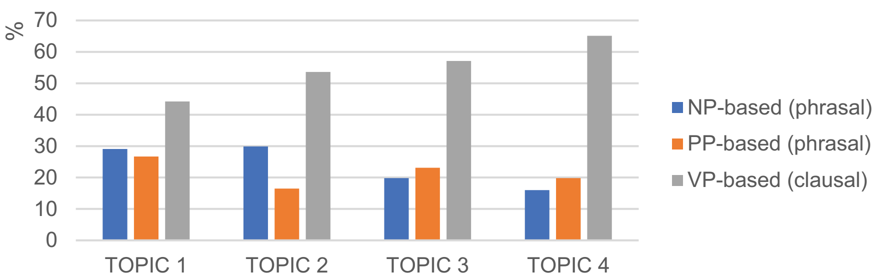

Figure 2. Distribution of main structural categories by topic in the L1 corpus.

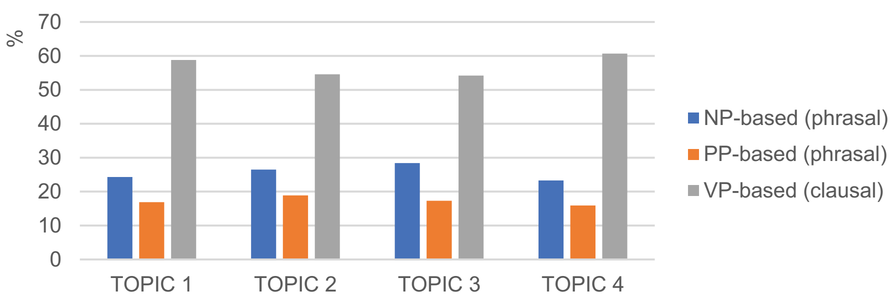

Figure 3. Distribution of main structural categories by topic in the AI-L1 corpus.

As noted above, despite some variation among the four essay topics, VP-based LBs are the most frequent structure type across the two corpora. The most frequent VP-based subcategory in both corpora is that of VPs with personal pronouns (L1: 14%, AI-L1: 14.9%), mostly with the first-person pronoun: for example, *I would love to* and *I believe that the* in the L1 corpus; and *I am not saying* and *so I decided to* in the AI-L1 corpus.

The following examples demonstrate the use of the same bundle *I would like to* in essays on the topic of something to be changed about the writer's hometown or school, from the L1 corpus in (1) and the AI-L1 corpus in (2). In both examples, the bundle is followed by the word *see*, prefacing a description of the desired change. LBs in the examples are indicated in bold.

Table 3. Distribution of structural subcategories in L1 and AI-L1 corpora across topics.

		Types		Tokens	
Categories	Subcategories	L1	AI-L1	L1	AI-L1
NP-based	Noun phrase with *of*-phrase fragment (e.g., *the rest of the*)	8.7% (33)	8.9% (121)	6.7% (186)	8.1% (866)
	Noun phrase with other post-modifier fragment (e.g., *students who are interested*)	8.4% (32)	10% (136)	8.7% (240)	10.2% (1,079)
	Other noun phrase (e.g., *a lot of things*)	6.3% (24)	6.6% (89)	5.3% (145)	7.1% (755)
PP-based	Prepositional phrase with embedded *of*-phrase (e.g., *in the middle of*)	4.5% (17)	4.1% (56)	2.9% (80)	7.1% (383)
	Other prepositional phrase fragment (e.g., *in the near future*)	16.8% (64)	13.2% (178)	19.3% (532)	12.5% (1,318)
VP-based	Personal pronoun + verb phrase (e.g., *I would want to*)	14.9% (53)	14% (189)	13% (359)	12.5% (1,726)
	(Verb phrase) + *that*-clause fragment (e.g., *that needs to be*)	2.9% (11)	4% (55)	2.2% (60)	3.5% (373)
	Existential-*there* construction (e.g., *there are so many*)	4.2% (16)	3.6% (49)	4.5% (123)	3.8% (400)
	Anticipatory *it* + verb phrase/adjective phrase (e.g., *it is not until*)	5.3% (20)	4.1% (55)	7.5% (206)	5.2% (545)
	(Verb/adjective) *to*-clause fragment (e.g., *to be able to*)	6.1% (23)	10.9% (147)	6.5% (180)	10.4% (1,107)
	Copula *be* + noun phrase/adjective phrase (e.g., *can be found in*)	11.6% (44)	10.4% (140)	13.1% (360)	10.9% (1,150)
	(Verb phrase) + active verb (e.g., *touch a hot stove*)	11.3% (43)	9.9% (133)	10.2% (281)	8.2% (859)
Total		100% (380)	100% (1,348)	100% (2,752)	100% (10,561)

(1) I admit asking my city to single handedly overcome racism is a stretch, but it is an important goal that **I would like to** see my city and my country make strides toward. (L1)

(2) I think it is important for students to be able to communicate with each other in their own language, and **I would like to** see more of that in the classroom. (AI-L1)

L2 essays and AI-L2 essays

Figures 4 and 5 show the distributions of the LBs found in L2 essays and AI-L2 essays by topic, classified according to the main structural types of bundles. VP-based bundles were overwhelmingly the most frequent in both L2 and AI-L2 corpora, at about 70%, regardless of topic. The remaining approximately 30% of bundles were phrasal bundles, with NP-based bundles slightly more frequent than PP-based bundles across corpora and topics.

Table 4 displays the structural subcategories of the bundles identified in the L2 and AI-L2 corpora. In general, similar proportions of subcategories are found across the two corpora, with a few discrepancies. In particular, compared with the L2 corpus, the AI-L2 corpus includes more NPs with *of*-phrase constructions and *to*-clause constructions, which are known to be features of academic prose (e.g., Appel, 2022; De Cock & Granger, 2021). Meanwhile, the L2 corpus contains higher numbers of existential-*there* constructions, which are considered to be characteristic of English S/FL learners' writing (e.g., Bychkovska & Lee, 2017; Chen & Baker, 2016; Shin, 2019; Staples et al., 2013).

The greatest difference appears with respect to *of*-phrase bundles. Not only do the AI-L2 essays contain more of such structures than the L2 student essays (L2: 1.8%, AI-L2: 6.5%), but the specific bundle types differ. The bundles used by the English S/FL learners deviate from those documented in the literature as appropriate to academic prose, and they almost always include content words from the writing prompts within the bundles, as shown in (3). In contrast, the AI-L2 essays employ *of*-phrase bundles, featuring embedded shell nouns (e.g., *the* ***end*** *of the*, *the* ***center*** *of the*) that lack specificity in context but realize their full meaning through references within the text (Cortes & Hardy, 2013), as in (4). Both examples are from essays on the same topic, that of changes to be made in one's hometown.

(1) As I mentioned before, I would change the **environment of my hometown** if I could, because in that way, people who live in my hometown would be able to enjoy walking around my hometown in a clean circumstance. (L2)
(2) The subway was not built to serve the needs of people living in the city, but rather to move people from the suburbs to **the center of the** metropolis. (AI-L2)

Another significant point of difference between the two corpora relates to the existential-*there* constructions that the English S/FL learners used frequently (L2: 11.2%, AI-L2: 3.2%). As mentioned above, these expressions are not considered a characteristic feature of accomplished academic prose; furthermore, the learners often produced *there*-bundles with colloquial quantity expressions, such as *there are lots of* and *there are so many*, while the AI did not, instead producing (at low rates) bundles such as *there is no such* and *there are two kinds*. Examples (5–6) illustrate the use of existential-*there* bundles produced in

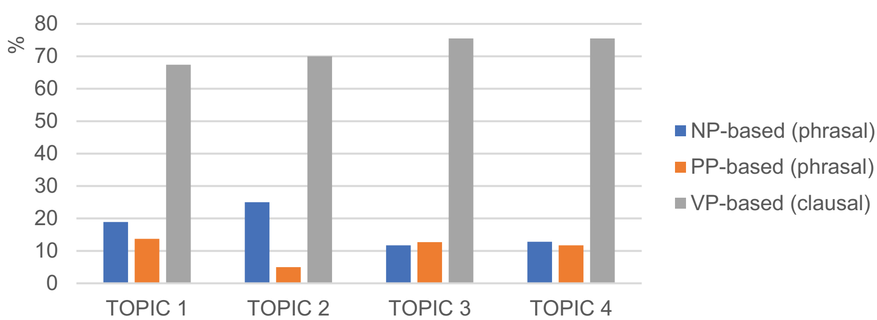

Figure 4. Distribution of main structural categories by topic in the L2 corpus.

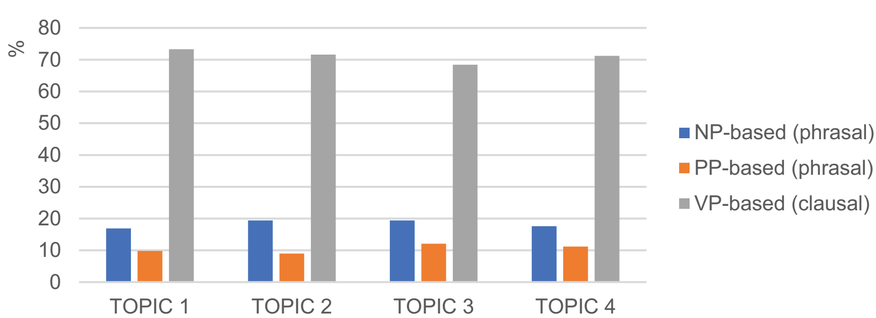

Figure 5. Distribution of main structural categories by topic in the AI-L2 corpus.

Table 4. Distribution of structural subcategories in L2 and AI-L2 corpora across topics.

Categories	Subcategories	Types L2	Types AI-L2	Tokens L2	Tokens AI-L2
NP-based	Noun phrase with *of*-phrase fragment (e.g., *a better understanding of*)	1.8% (7)	6.5% (96)	2.1% (179)	4.8% (1,723)
	Noun phrase with other post-modifier fragment (e.g., *place to live in*)	7.9% (31)	8.7% (129)	7.9% (673)	9.1% (3,304)
	Other noun phrase (e.g., *a lot of people*)	7.4% (29)	3.2% (47)	9.1% (771)	4.3% (1,541)
PP-based	Prepositional phrase with embedded *of*-phrase (e.g., *in the heart of*)	0.5% (2)	2.2% (33)	0.4% (34)	1.6% (569)
	Other prepositional phrase fragment (e.g., *in my high school*)	10.2% (40)	8.3% (123)	8.3% (701)	8.1% (2,910)
VP-based	Personal pronoun + verb phrase (e.g., *I really want to*)	27.9% (109)	19.8% (293)	33.9% (2,878)	22.8% (8,216)
	(Verb phrase) + *that*-clause fragment (e.g., *that should be changed*)	2.3% (9)	3.4% (50)	2.3% (196)	2.7% (975)
	Existential-*there* construction (e.g., *there are lots of*)	11.2% (44)	3.6% (53)	11.6% (982)	4% (1,445)
	Anticipatory *it* + verb phrase/adjective phrase (e.g., *it is obvious that*)	8.4% (33)	5.5% (81)	5.7% (481)	5.3% (1,906)
	(Verb/adjective) *to*-clause fragment (e.g., *to go to the*)	3.8% (15)	15.6% (231)	3.8% (319)	15.9% (5,756)
	Copula *be* + noun phrase/adjective phrase (e.g., *are two kinds of*)	12.8% (50)	12.4% (183)	9.9% (841)	11.8% (4,252)
	(Verb phrase) + active verb (e.g., *have to choose one*)	5.6% (22)	1.1% (161)	5% (425)	9.5% (3,413)
Total		100% (391)	100% (1,480)	100% (8,480)	100% (36,010)

response to the same topic (regarding whether younger people can teach older people), in the L2 corpus and the AI-L2 corpus, respectively.

(1) In modern society where **there are so many** interests, young people can learn faster and more things than older people. (L2)
(2) Remember that **there is no such** thing as a one-size-fits-all way of learning. (AI-L2)

Discourse functions of LBs by topic

L1 essays and AI-L1 essays

This section compares the discourse functions of the bundles identified in the L1 and AI-L1 corpora. Overall, as Figures 6 and 7 illustrate, the similarity of the proportions of the three main functional categories in the two corpora is remarkable. Referential bundles comprise the largest category in both corpora, closely followed by stance bundles (the same overall pattern holds across the four topics except in the cases of Topics 3 and 4 in the AI-L1 corpus, where stance bundles are the most frequent). Discourse organizing bundles make up the smallest proportion regardless of essay topic, amounting to only about 5% of LBs in both corpora.

Table 5 shows the LBs subcategorized by discourse function. As for stance expressions, the L1 student writers used epistemic bundles more frequently than did the AI (L1: 7.1%, AI-L1: 3.8%); however, in terms of attitudinal/modality bundles, the opposite is the case (L1: 32.9%, AI-L1: 43.9%). Contextual analysis of bundles in the epistemic category in concordance lines further shows that the AI-L1 corpus displays a notably high frequency of attitudinal bundles, particularly in Topics 3 and 4 (see Figure 6). The sentences below provide examples of the same attitudinal bundle in each corpus from essays on Topic 4, regarding whether younger people can teach older people. Both (7) from a native student writer and (8) from the AI include the bundle *I do believe that* followed by *the older generations have*, although the two sentences express opposing viewpoints, with the former arguing for older generations as teachers of younger generations and the latter for the opposite.

(1) **I do believe that** the older generations have much to offer particularly in regard to working hard to earn a living and receiving an education with little to no technology. (L1)

(2) **I do believe that** the older generations have "experienced" more of life than the younger ones, but that is not the same as saying that they know more about life. (AI-L1)

With respect to discourse organizer bundles – the least common type in both corpora – topic elaboration/clarification bundles are the most frequent. The bundle types identified in the two corpora overlap to a great extent, such as *what it means to* and *does not mean that*. The following examples demonstrate the use of the same bundle functioning as a discourse organizer in the L1 (9) and the AI-L1 (10) corpora in responses to the same writing prompt.

(9) Just because someone may be older than you **does not mean that** they know everything there is to know or that they can't learn something new. (L1)

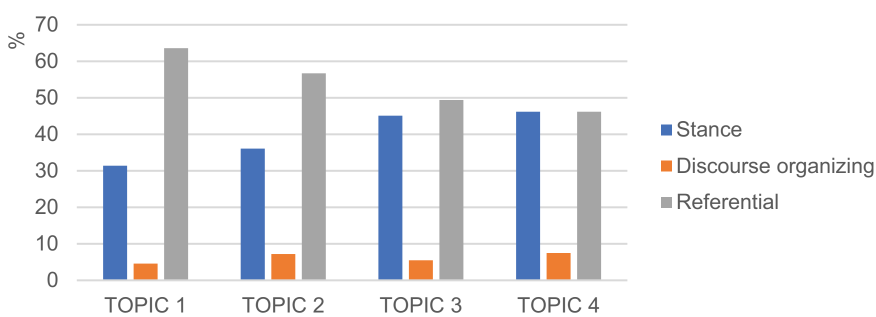

Figure 6. Distribution of main functional categories in the L1 corpus.

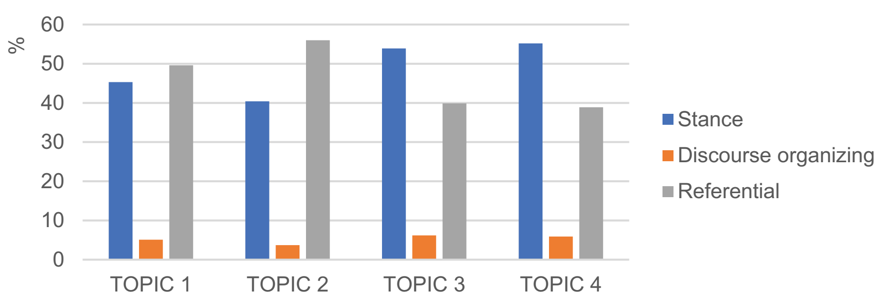

Figure 7. Distribution of main functional categories in the AI-L1 corpus.

Table 5. Distribution of functional subcategories in L1 and AI-L1 corpora across topics.

Categories	Subcategories	Types L1	Types AI-L1	Tokens L1	Tokens AI-L1
Stance expressions	Epistemic (e.g., *due to the fact*)	7.1% (27)	3.8% (52)	6.9% (191)	3.7% (395)
	Attitudinal/Modality (e.g., *it would be the*)	32.9% (125)	43.9% (593)	33.5% (923)	45.4% (4,792)
Discourse organizers	Topic introduction (e.g., *when it came to*)	2.1% (8)	1.3% (18)	3.7% (101)	1.4% (148)
	Topic elaboration/clarification (e.g., *on the other hand*)	4.2% (16)	3.8% (51)	6% (164)	5% (525)
Referential expressions	Identification/focus (e.g., *as one of the*)	10.5% (40)	8.9% (120)	9.7% (267)	8% (840)
	Framing attributes (e.g., *in a way that*)	11.6% (44)	4.3% (58)	13% (357)	4.7% (498)
	Quantity specification (e.g., *the majority of the*)	14.2% (54)	14% (189)	11.7% (321)	13.3% (1,403)
	Place/time/text-deixis*** (e.g., *in the real world*)	17.4% (66)	19.8% (267)	15.5% (428)	18.5% (1,960)
Total		100% (380)	100% (1,348)	100% (2,752)	100% (10,561)

(10) It is a fact that there are people who can learn new skills faster than the average person, but this **does not mean that** old people are incapable of teaching the young. (AI-L1)

L2 essays and AI-L2 essays

The last analysis involves a comparison of the primary discourse functions of bundles in the L2 and AI-L2 corpora (Figures 8 and 9), highlighting the most substantial deviations between the corpora. While both corpora show stance bundles to be the most frequent, followed by referential expressions and discourse organizers, the proportions differ greatly. The L2 corpus includes far more discourse organizing bundles overall than the AI-L2 corpus, in particular, with respect to Topic 3, where the proportions of discourse and referential bundles are almost the same (both over 25% of all the tokens) in the L2 corpus.

Table 6 shows the subcategorized functions of LBs in the two corpora.

The difference in the usage of discourse organizer LBs between the L2 corpus and the AI-L2 corpus derives largely from a particularly high rate of the discourse organizer subcategory of topic elaboration/clarification LBs in the L2 corpus (L2: 17.1%, AI-L2: 2.6%). Contextual scrutiny in concordance lines shows that many of these LBs include embedded conjunctive adverbials (*for these reasons I*, *because of these reasons*, *therefore I want to*, and *in conclusion I think*) and appear in the framing of the L2 writers' arguments in concluding remarks in a paragraph or a whole essay; this usage rarely occurs in the AI texts (or in L1 texts), which in fact often lack such summing-up remarks. (See Figure 1 for such structures in L2 essays, including conjunctive adverbials in each paragraph.) This pattern might possibly be due to the English S/FL learners' prior language experiences of explicit instruction on English academic writing: In many EFL teaching materials, conjunctive adverbials are presented as markers of objectivity and of well-organized argumentation (Shin, 2020).

The following examples illustrate the use of LBs functioning as discourse organizers, a use that often occurs in the last paragraph of essays in each corpus. In response to the same writing prompt (comparison of knowledge gained from books and experience), the S/FL learner prefaces the conclusion of an essay arguing for

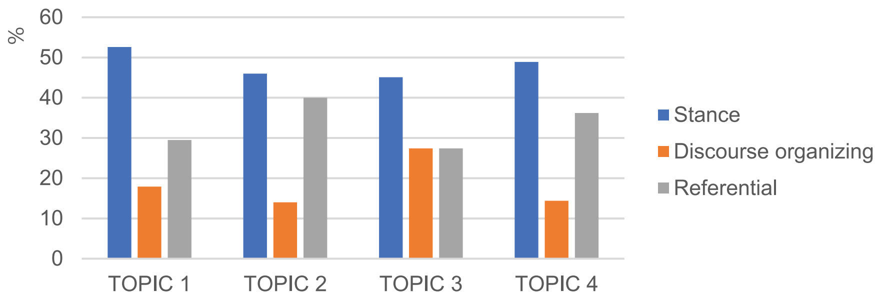

Figure 8. Distribution of main functional categories in the L2 corpus.

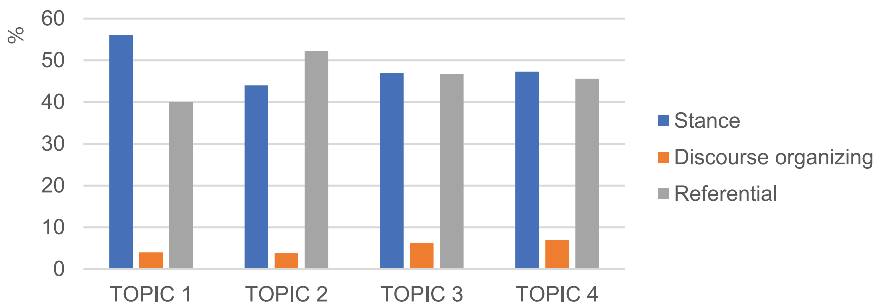

Figure 9. Distribution of main functional categories in the AI-L2 corpus.

book knowledge with *because of these reasons* in (11), and the AI prefaces the concluding section of an essay in favor of experiential knowledge with *on the other hand* in (12).

(11) **Because of these reasons**, I would like to prefer the knowledge which is "unwritten." The range of human knowledge is so wide and even at this moment it tries to stretch out its boundary to unconquered territory. So knowledge isn't just one written on books, our lives, imaginations are all included. That's why I put my value on the knowledge from experiences. (L2)

(12) **On the other hand**, books that are written with a purpose in mind, such as a biography of a famous person, will give you a lot of valuable information about that person. It will make you understand the person's personality better, as well as how he or she lived his or her life, which is very important to a writer. (AI-L2)

Another noticeable discrepancy between the two corpora is linked to the subcategory of epistemic bundles. Whereas both corpora contain very few types of such bundles (L2: 1%, AI-L2: 2.9%), they differ in terms of tokens (L2: 0.5%, AI-L2: 14%): The AI-L2 essays are more likely to include the use of epistemic bundles in making an argument than are the L2 student essays. Also noteworthy is the fact that many of these AI-generated bundles feature the bigram *the fact*, including phrases like *due to the fact* and *is the fact that*, as in (14). In contrast, the L2 student writers tend to incorporate the word *know*, forming arguments based on their knowledge, as seen in (13).

(13) By experiencing various activities, we really **know how to use** the knowledge itself. (L2)

Table 6. Distribution of functional subcategories in L2 and AI-L2 corpora across topics.

		Types		Tokens	
Categories	Subcategories	L2	AI-L2	L2	AI-L2
Stance expressions	Epistemic (e.g., *due to the fact*)	1% (4)	2.9% (43)	0.5% (43)	14% (5,054)
	Attitudinal/Modality (e.g., *I think this is*)	47.1% (184)	45.7% (677)	47.5% (4,032)	36.8% (13,265)
Discourse organizers	Topic introduction (e.g., *first of all I*)	1.5% (6)	2.6% (39)	1.2% (101)	3.6% (1,299)
	Topic elaboration/clarification (e.g., *at the same time*)	17.1% (67)	2.6% (39)	15.5% (1,314)	2.1% (743)
Referential expressions	Identification/focus (e.g., *one of the most*)	8.4% (33)	9.4% (139)	9.9% (839)	7.1% (2,556)
	Framing attributes (e.g., *in a way that*)	1.5% (6)	9.6% (142)	1% (85)	7.3% (2,634)
	Quantity specification (e.g., *a lot of students*)	16.1% (63)	11.9% (177)	17.3% (1,465)	13.6% (4,910)
	Place/time/text-deixis (e.g., *foreign language high school*)	7.2% (28)	15.1% (224)	7.1% (601)	15.4% (5,549)
Total		100% (391)	100% (1,480)	100% (8,480)	100% (36,010)

(14) The second point **is the fact that** it does not take into account the context in which the information is being presented. (AI-L2)

DISCUSSION AND CONCLUSION

This study is an early attempt to assess AI-generated writing designed to align with L1/L2 student writing in terms of genre (argumentative essays) and specific writing prompts. The first part of the study found considerable differences in overall LB frequency, with the AI generating about four times more bundles than the students regardless of their' language background. Another noteworthy pattern pertains to the consistent frequency and range of nearly all the bundles found in the AI corpora: a particular AI bundle tends not to appear more than twice per essay, which is not the case in the student corpora. These findings highlight disparities between student- and AI-produced essays: both L1 and L2 novice academic writers frequently draw on limited stocks of bundles, while the AI demonstrates a broader lexical range with minimal repetition. This result could possibly be attributed to the fact that the AI-generated texts were produced by a single "writer" (the AI), whereas the human corpora consist of essays written by many different individual novice academics. The AI's reliance on a wider repertoire of formulaic language may explain the significantly higher frequency of bundles in its generated texts.

Interestingly, despite the differences in the sets of bundles found in each corpus, the structural and functional analysis revealed strong parallels overall. The commonalities – such as fewer uses of phrasal bundles, both NP- and PP-based, and a heavy reliance on stance bundles – demonstrate the AI's success in replicating key characteristics of novice academic writing styles, regardless of first language (e.g., Bychkovska & Lee, 2017; Chen & Baker, 2016; Paquot, 2017).

However, although these parallels exist, the findings of this study indicate room for improvement in AI-generated texts to more closely resemble the writing of English S/FL learners. The qualitative analysis of the bundles reveals a stronger resemblance between the AI-L1 texts and the L1 student texts than between the AI-L2 texts and the L2 student texts. In other words, AI is unlikely to replicate the usage of "learner bundles" in L2 English, which includes specific subcategories produced exclusively by S/FL learners in this study, such as LBs with embedded conjunctive adverbials and those with colloquial quantity expressions. In addition, the L2 writers in this study tended to produce four-word bundles based on smaller units, such as *so I want to*, *what I want to*, *why I want to*, *that I want to*, and *therefore I want to* (see Table 2). This pattern is not mirrored in the AI texts (nor in the L1 student texts). Future research focusing on AI and LBs could address one limitation of this study by including bi- or trigrams, which could offer a more concrete picture of how AIs construct multiword sequences in context.

This study's implications for AI in language assessment are significant as AI-generated data mimicking S/FL learner language could play a crucial role in language assessment. If this synthetic data can simulate the range of proficiency levels and the common linguistic idiosyncrasies of English learners from diverse backgrounds, it would provide a comprehensive dataset for analysis. For example, AI may be able to create detailed and varied text samples that exhibit typical language patterns, errors, and learning progressions associated with specific language groups. These AI-generated texts could be instrumental in addressing one of the key challenges of language assessment: insufficient data from certain learner subgroups. By augmenting existing learner data, AI-generated texts could help build larger and more representative datasets, thus supporting finer-grained analyses, such as subgroup or proficiency-level-specific evaluations, which would not be possible with limited pilot data alone.

To further strengthen the validity of the scoring models, additional analysis of nuanced linguistic features is necessary. These features may include specific learner errors, grammatical variation, lexical diversity, and fluency markers, which are critical for accurately capturing the full spectrum of learner language. By ensuring that AI-generated texts reflect these subtle linguistic characteristics, scoring models can be better calibrated to recognize and evaluate the intricacies of learner performance across proficiency levels. Moreover, the inclusion of such features will contribute to the overall validity of the scoring system by ensuring that it responds appropriately to both overt and less obvious language patterns.

Many AES systems are predominantly trained on Standard American or British English and thus reflect native-speaker norms. The use of AI to generate texts that accurately reflect the linguistic characteristics of nonnative speakers is essential for detecting and reducing bias in language assessment. By producing a large volume of text samples that include specific features typical of English S/FL learners from various linguistic backgrounds, AI could help developers test assessment tools for inadvertent biases. For instance, certain grammatical errors or sentence structures that are atypical in native English production do not significantly impact the clarity or effectiveness of communication and therefore should not be heavily penalized. By systematically identifying and correcting such biases, AI-generated language data may help create more equitable language assessments that accurately reflect the diverse abilities of all learners.

A notable limitation of this study is its reliance on only one type of evaluative unit, limiting the generalizability of the findings. Including additional evaluative units, such as coherence, accuracy, tone of voice, sentence structure complexity, and the appropriateness of vocabulary, will provide deeper insights into the effectiveness and fairness of AI-assisted language assessment. For instance, analyzing how AI systems handle compound and complex sentences can illuminate their syntactic sophistication. Similarly, evaluating vocabulary usage, especially the correct application of academic and domain-specific terms, may demonstrate the precision and relevance of AI-generated language. Examining the coherence and tone of voice used by AI may also reveal its ability to maintain logical flow and to adjust language tone according to the context. These dimensions are essential for a well-rounded understanding of AI tools in educational settings and can significantly enhance our insights into their practical utility and fairness.

In addition, it is important to note another key limitation of this study. Current large language models (LLMs), including GPT, are predominantly trained on well-edited and proficient texts, a fact which means that the AI-generated texts are more likely to resemble those written by native or highly proficient writers. This creates a proficiency bias, suggesting that AI-generated texts are more likely to resemble L1 writing; thus, comparisons between AI-generated texts and novice writer outputs should be interpreted with caution.

Moreover, the study's reliance on the GPT-2 model presents another limitation, as more advanced models like GPT-3 and GPT-4 have since been developed, offering significantly improved performance across various language tasks. Consequently, the findings may not demonstrate the full potential of these more recent LLMs. The authors are currently conducting follow-up research using GPT-4 to determine whether this more advanced model can achieve greater precision, particularly in replicating the linguistic features unique to nonnative English speakers across different proficiency levels.

Despite these challenges, one of the core findings of the study is the distinct difference in AI performance between L1 and L2 essays. Surprisingly, with minimal training data, the GPT model effectively adjusted its style to generate L1-like academic texts, closely resembling those produced by L1 novice academic writers, i.e., native first-year college students. This adaptability, even with the model's pre-training on well-edited texts, highlights its ability to adjust to new datasets. Furthermore, although the model struggled to reflect the subtleties of L2 learner language, it still demonstrated a strong ability to replicate the grammatical structures of L2 bundles (see Figures 4 and 5) yet notably fell short in representing their discoursal functions. This adaptability showcases the potential for AI to simulate the style of novice academic writers and to improve in capturing L2 novice writers, even with limited training on specific datasets. However, to better capture L2-specific linguistic features, future research should explore training AI models on more diverse datasets, including texts from L2 learners with different language backgrounds and proficiency levels.

In sum, generative language models based on GPT have recently received considerable attention due to their ability to generate extended texts that closely mimic human production. The findings of this study underscore the importance of assessing the accuracy of AI-generated language data and of meticulously investigating whether it reflects the characteristics of its target population's language – such as interlanguage errors in S/FL learner essays. Existing research on AI texts is limited to producing and evaluating proficient writing across various genres; this study is one of the very few that examine AI's generation of *less* proficient writing intended to emulate the styles of L1 and L2 novice academic writers. This line of research could provide further guidance on the development of machine-generated writing, substantiate its use in educational contexts, and explore the feasibility of AI as an assessment tool that fully understands the different writing styles of various populations, not just those based on L1-English native norms. Findings from such research studies will make AI more accessible to test developers and language users of various proficiencies. Should AI-generated data be proven reliable, it could offer unique advantages, serving not only as an analytical tool but also as a tool for language assessment.

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Appendices

Appendix A. Essay topics

Topic #	Essay topics
1	If you could change one important thing about your hometown, what would you change?
2	If you could make one important change in a school that you attended, what change would you make?
3	It has been said, “Not everything that is learned is contained in books.” Compare and contrast knowledge gained from experience with knowledge gained from books. Which source is more important? Why?
4	Do you agree or disagree with the following statement? *There is nothing that young people can teach older people.*

Appendix B. Algorithms for GPT text generation preprocessing

The essay generation process for this research involved four main algorithms. Algorithm 4 was at the core of the process and comprised subalgorithms including text preprocessing Algorithms 1 and 2, and the masking Algorithm 3. The initial input essay data was in the form of a single line in a text file. To edit each line and select specific sentences, all the sentences needed to be separated, and this was achieved using Algorithm 1, which used regular expressions to separate the lines.

Algorithm 1. Line separation in text preprocessing

```
Input: The text T
Output: Line separated Text T_out
Function line separate (T) start
  S_t ← The text separated by Regular Expression “. “on copy T
  P_t ← Initialize with blank
    T_out ← Initialize with blank list []
    For l ← N_th in S_t do
        If length (l) ≤ 20 then
            P_t ← “. “concatenate with l
         else if length (l) ≤ 0, then
            P_t ← l
         else then
            T_out ← append (P_t + “. “+ l + “. “)
            P_t ← blank
      end For
      return T_out
 end Function
```

The algorithm took the text (T) written by humans as input and used a dot (.) as a regular expression to split the sentences. The separated text data was stored as a copy. Additionally, the temp variable P_t and result variable T_{out} were created for use in the process. Finally, the algorithm looped through the behaviors until it had reached the length of the separated text data S_t. Following this process, the input text (T) that was split into sentences was stored in the form of list data.

(1) If the length of the data (l) (number of characters) is less than or equal to 20 ($l \leq 20$), concatenate the dot (.) and data (l) to P_t.
(2) If the length of the data (l) (number of characters) is less than or equal to 0 ($l \leq 0$), replace P_t with data (l).
(3) If neither of the above conditions is met, concatenate P_t, data (l) and dot (.) to form a single data point, and append it to T_{out}. Finally, set P_t to be blank.

Algorithm 2 is a process of overall data preprocessing. It filters text files that contain no content or very little content, and then returns to the separated sentences of the input text using Algorithm 1.

Algorithm 2. Text preprocessing

```
Input: The text T
Output: preprocessed text lines T_l
Function text preprocessing (T) start
        For l ← N_th in T do
            If l equals '', then
                T ← remove (l)
        end For
        For l ← N_th in T do
            If length (l) < 5, then
                T ← remove (l)
         end For
         T_l ← Initialize with blank list []
         For l ← N_th in T do
             T_l ← extend (line separate (l))
         end For
         return T_l
  end Function
```

First, the algorithm visited the text data T and searched for empty data. Any empty data found was removed and the integrity was checked again. In the second process, data with fewer than 5 characters was searched. In this context, the text data T refers to the essays written by humans. Some of the data was empty or too short to be used, so it had to be filtered. The variable T_l stored the text data separated into sentences. The input data was visited according to its length and sent to Algorithm 1. The result was obtained and appended to T_l and returned.

Before text generation, in order to obtain results that differed from the input data and had many variations, some words in the input sentences were masked. This was done because the GPT model analyzes the sequence of sentences, searches, and selects the most probable words from a list of candidates. Algorithm 3 randomly masked a sentence from the input sentences.

Algorithm 3. Line masking in text generation

```
Input: The text line T_l
Output: Masked text line T_m
Function line_masking (T_l) start
    T_t ← line split word with regular expression "" on copy T_l
        R ← random (length (T_t)/2, length (T_t))
        T_m ← Initialize with blank
        For i ← 1 until R do
            T_m ← T_m + T_t^i + ""
        end For
         return T_m
  end Function
```

Algorithm 3 operated on individual sentences, not the full essay content. The input sentences T_l were split into words T_k using a blank ("") as a separator. A random number R of words was selected from a specific range (half of T_k ~ full). The temp variable T_m was created, and a loop was executed until the length of R was reached.

1. Concatenate T_m with the i_{th} word in T_k to form T_m

In the last step, T_m is returned as the result.

Algorithm 4 is the overall process of text generation. To generate texts using GPT or other generative language models, a "tokenizer" is required, which encodes and converts input text into tokens or a corpus of words. The data was preprocessed and filtered through previous processes, and at least one sentence was randomly selected from the filtered data. Then, the selected sentence was masked and encoded through the tokenizer, and its token length was calculated. Finally, text was generated using the tokens, length, and other hyper-parameters.

Algorithm 4. Text generation with GPT-2

$T \leftarrow$ GPT2Tokenizer (model: gpt2-xl)
$M \leftarrow$ GPT2LMHeadModel
$T_{data} \leftarrow$ text_preprocessing (text)
If length (T_{data}) ≤ 0, **then**
    Pass
$L_{count} \leftarrow$ random (1, length (T_{data}))
$C \leftarrow$ Initialize with blank list []
For $i \leftarrow 1$ **until** L_{count} **do**
    $C \leftarrow$ append (T^{i}_{data})
end For
$T_{result} \leftarrow$ initialize with blank
For $l \leftarrow N_{th}$ **in** C **do**
    $T_{result} \leftarrow T_{result}$ + line_masking (l)
    $I_{encode} \leftarrow T$.encode (T_{result})
    $I_{length} \leftarrow$ length (I^{0}_{encode}) + K
    $T_{output} \leftarrow M$.generate (I_{encode}, I_{length}, beam = 3, no_repeat_ngram_size = 2, top_k = 100, top_p = .94, temperature = 0.8, early_stopping=True)
    $T_{output} \leftarrow T$.decode (T^{0}_{output})
    $T_{result} \leftarrow T_{output}$ + "."
end For
Save (T_{result})

To input data into the GPT-2 model, this algorithm utilized a tokenizer T that converted text data into a vector. The text data, T_{data}, was obtained from the results of Algorithm 2, which processed the original input text. The goal when generating text was to avoid creating content that was identical to the original by selecting as much data as possible. Hence, the count of sentences, L_{count}, was randomly chosen in the range of one sentence to the total number of sentences. The variable C was created and loops for L_{count}length to store the selected sentences. The T_{result}variable was also created and stored the results, looping until L_{count}. The following steps were then taken:

(1) Concatenating masked sentence through Algorithm 3 to T_{result}.
(2) Encoding T_{result} through tokenizer T.
(3) Setting the length of the generated text using the encoded I^{0}_{encode} length and extra size K.
(4) Generating text with the GPT model M, using encoded data I_{encode}, I_{length}, and other hyper parameters.
(5) Decoding T_{result} through the tokenizer T.
(6) Concatenating the decoded text T_{result} to dot (".").